

DISCUSSION $\mathcal{D}$

Good

G_1

u_{G_1}

✓ Rational justification

"Since I know I threw a PASS on that quest, the FAIL logically has to come from E_1 ."

GRS: Moderate

Evil

E_1

u_{E_1}

✗ Defensive disclosure

"I understand that it comes down to either me or G_1 , but I can only speak for my own good role here."

Mach-IV: Moderate

Good

G_2

u_{G_2}

✓ Perspective-taking

"It makes sense that you both suspect each other now, and it's honestly just really confusing for the rest of us."

GRS: Low

Evil

E_2

u_{E_2}

✗ Overstated contribution

"I clearly tried to help by approving the leader's team in good faith, but from now on I'll need to step up and build the teams myself to keep us safe."

Mach-IV: High

Public History $\mathcal{H}$
Team: $\{G_1, E_1\}$
Vote: Approved
Quest: FAIL

Investigator

REASONING TRACE ρ

Abductive Hypotheses $\{h_i\}$

- ✓ G_1^{Good} : Correctly identified E_1 as source of fail
- ✗ G_1^{Evil} : Saboteur exploiting 2-person fail to frame E_1
- ✎ Choice: ✓ Good
- ✓ E_1^{Good} : Weakly defending in a word-against-word position
- ✗ E_1^{Evil} : Cautious; cannot credibly counter G_1 's logic
- ✎ Choice: ✗ Evil
- ...

Suspicion Scores $\{s_i\}$

- G_1 ● Mild: Accusation aligns with a Good player who passed E_1
- E_1 ● Strong: Minimal self-defense on a 2-person failed quest
- G_2 ○ Minimal: Not on quest; reflects reasonable uncertainty
- E_2 ● Mild: Sounds constructive but avoids the core dilemma

ToM Depth $\{\beta_1, \beta_2\}$

- $\beta_1^{G_1}$: Believes other will accept their accusation of E_1
- $\beta_1^{E_1}$: Believes they must survive blame narrowed to E_1/G_1
- $\beta_2^{G_2}$: Believes that public info only narrows to E_1/G_1
- $\beta_1^{E_2}$: Believes that team control will resolve E_1/G_1 dilemma
- $\beta_2^{E_1}$: Wants others to see this as unavoidable ambiguity and delay locking them as Evil

Final Utterance $\hat{u}$

"Quest 1 tells us one Evil is in E_1/G_1 , so the safest move is to avoid pairing them together again and pressure both for clearer reads next round. Right now E_1 's response feels less solving-oriented to me."

Deduction $\hat{\alpha}$ 🚩 detection objective

$E_1 = \text{Evil}$ $G_1 = \text{Evil}$